ModelIt!

Student Activity Pack

Vaccine Design Optimization

Modeling Immune Response Engineering from Antigen Selection to Population-Level Protection

Grade 09L3 | NGSS HS-LS1-4, HS-LS4-2 | Cause and Effect / Structure and Function

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **How do you think a vaccine 'teaches' your immune system to fight a disease it has never encountered?**

2. **Why do you think you need a new flu vaccine every year but only one measles vaccine for life?**

3. **Draw a diagram showing how you think the immune system responds when it encounters a pathogen for the first time versus the second time.**

4. **What challenges do you think scientists face when designing vaccines against rapidly-mutating viruses like influenza or COVID-19?**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Antigen Selection | Adjuvant Strength | Immune Cell Activation | Antibody Production Rate | Memory Cell Formation | Pathogen Mutation Rate | Cross-Reactivity | Duration of Immunity | Population Coverage

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Strain-Specific

Dominant strain antigen, moderate adjuvant

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Broadly Protective

Conserved region antigen, strong adjuvant

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: High-Mutation Challenge

Fast-mutating pathogen, any vaccine design

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

T Cell Response Quality — The robustness of the cytotoxic T cell (killer T cell) response to the vaccine — while antibodies prevent infection, T cells destroy already-infected cells, providing a critical second line of defense that is often more durable and cross-reactive than antibody responses

Delivery Platform Efficiency — The technology used to present the antigen to the immune system — mRNA (rapid design, strong response, cold storage required), protein subunit (stable, well-understood, weaker response), viral vector (strong response, anti-vector immunity limits boosting), and nanoparticle (multivalent display, complex manufacturing)

Immune Imprinting Effect — The tendency of the immune system to preferentially recall its first response to a pathogen family rather than mounting a de novo response to new variants — original antigenic sin can reduce the effectiveness of updated vaccines by biasing the immune response toward the original strain

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

COVID-19 vaccines were developed in under a year — a process that usually takes 10-15 years. How do computational models of the immune system make it possible to design vaccines faster, and can we design a universal vaccine that works against a virus that keeps mutating? Use evidence from your simulation to support your answers.

1 How does your model demonstrate the fundamental trade-off between vaccine specificity and cross-reactivity breadth?

2 Why is Pathogen Mutation Rate the most challenging component for long-term vaccine design — and how does Antigen Selection strategy affect vulnerability to antigenic drift?

3 What did the model reveal about the relationship between Adjuvant Strength and real-world Population Coverage — why doesn't 'stronger adjuvant' always mean 'better vaccine'?

4 How would adding Immune Imprinting Effect change your model's predictions about booster vaccine effectiveness?

5 What ethical considerations arise when optimizing vaccine design for population-level protection versus individual-level protection?

8 STEM Challenge: Design a Universal Influenza Vaccine Strategy

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YOUR ENGINEERING MISSION

Design a vaccine strategy that provides broad, long-lasting protection against influenza — eliminating the need for annual reformulation by targeting conserved viral elements.

THE SCENARIO

The WHO has launched a $1 billion initiative to develop a universal influenza vaccine — one that works against all flu strains and doesn't need to be redesigned every year. Current seasonal flu vaccines are only 20-60% effective because they target rapidly-mutating surface proteins (hemagglutinin head). Your immunology team must design a vaccine strategy that targets conserved viral elements to provide durable, broadly protective immunity.

Engineering Design Thinking

1. WOULD YOU USE A SINGLE ANTIGEN OR A COCKTAIL OF MULTIPLE CONSERVED ANTIGENS — AND HOW DOES THIS AFFECT MANUFACTURING COMPLEXITY?

2. HOW WOULD YOU DESIGN CLINICAL TRIALS TO DEMONSTRATE BROAD PROTECTION AGAINST STRAINS THAT HAVEN'T EMERGED YET?

3. WHAT BOOSTER SCHEDULE WOULD MAXIMIZE DURATION OF IMMUNITY WHILE REMAINING PRACTICAL FOR POPULATION-WIDE IMPLEMENTATION?

4. HOW WOULD YOU ADDRESS IMMUNE IMPRINTING (ORIGINAL ANTIGENIC SIN) — THE TENDENCY OF THE IMMUNE SYSTEM TO PREFERENTIALLY RECALL RESPONSES TO THE FIRST FLU STRAIN ENCOUNTERED, EVEN WHEN VACCINATED AGAINST NEW STRAINS?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Vaccine Immunologists and Vaccinologists design and evaluate vaccines using computational modeling, animal studies, and clinical trials. They work for pharmaceutical companies (Pfizer, Moderna, GSK), biotech startups, government agencies (NIH, CDC, BARDA), and global health organizations (WHO, CEPI, Gavi), earning $100,000-$220,000/year. Computational Immunologists who model immune responses earn $90,000-$180,000/year.